



Technical Appendix

Avatar Tethering Methodology

Supporting Responsible AI Integration in Clinical Care

Submitted in response to:

HHS Request for Information — Accelerating the Adoption and Use of Artificial Intelligence in Clinical Care

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I. Overview

Avatar Tethering is a patented digital interaction methodology designed to enhance AI-supported healthcare engagement by contextually linking (“tethering”) relevant entities within a structured care environment. The system operates within an AI intermediary platform and is intended to:

- Improve care navigation
- Support workforce augmentation
- Deliver contextually relevant health information
- Facilitate escalation to licensed providers
- Sustain rural healthcare institutions through ethical partnership models

Avatar Tethering does not replace licensed medical professionals. Instead, it augments traditional healthcare delivery with by strengthening clinical and educational coordination between patients, providers, institutions, approved partners and Ai.

II. Core Concept

An “avatar” within this framework represents a digital identity or node corresponding to:

- A patient
- A licensed healthcare provider
- A healthcare institution (hospital, clinic, health department)
- A care navigator or nurse
- A sponsor or educational partner
- A pharmaceutical or therapeutic information source

“Tethering” refers to the structured digital linking of these avatars within a secure AI-mediated interaction based on user needs, clinical context, and workflow stage.

The result is a contextual, dynamic care environment in which interactions are coordinated, relevant, and responsive.

III. System Architecture

1. AI Intermediary Layer

The AI functions as a triage and workflow augmentation system that:

- Collects patient-reported symptoms or inquiries
- Performs structured triage
- Provides educational or preventive guidance
- Escalates to licensed providers when appropriate

The AI does not independently diagnose or prescribe without licensed clinician oversight.

2. Avatar Network Layer

The platform organizes participants into a structured digital ecosystem:

- **Patient Avatar** — Represents the individual seeking care.
- **Provider Avatar** — Represents licensed clinicians available for escalation.
- **Institutional Avatar** — Represents rural hospitals, clinics, or health departments.
- **Partner/Sponsor Avatar** — Represents approved educational or pharmaceutical partners delivering relevant information within regulatory boundaries.

The AI algorithm determines which avatars are tethered during the interaction based on contextual relevance.

3. Contextual Relevance Engine

Avatar Tethering operates on:

- Symptom-based triggers
- Preventive care indicators
- Chronic disease markers
- Workflow stage (triage, escalation, follow-up)
- Geographic routing logic (in-state provider prioritization)

This ensures that any linked educational, clinical, or sponsor-based content remains relevant to the patient's immediate needs.

IV. Regulatory Positioning

Avatar Tethering is structured as a **non-autonomous AI workflow coordination methodology** rather than a medical device.

Key characteristics include:

- Licensed provider oversight for diagnosis and prescribing
- Transparent disclosure of partner or sponsor participation
- Compliance with HIPAA and applicable privacy standards
- Data minimization principles

- Audit trails for interaction review

The methodology aligns with risk-proportionate regulatory frameworks for AI clinical support systems.

V. Revenue Sustainability Model

A distinguishing feature of Avatar Tethering is its ability to support sustainable rural healthcare funding without increasing patient costs.

The model allows:

- Contextual educational sponsorship placements
- Ethical pharmaceutical information delivery
- Institutional partnership engagement
- Telehealth escalation routing

Revenue derived from these partnerships can be shared among participating rural healthcare institutions under state governance structures.

This approach provides:

- Supplemental funding for rural hospitals
- Incentives for local provider participation
- Reduced reliance on unstable traditional tax bases

Importantly, content delivery is relevance-based rather than indiscriminate advertising, supporting patient-centered engagement.

VI. Alignment with HHS Objectives

Avatar Tethering directly supports federal priorities in:

1. Reducing Wasteful Utilization

- Improved triage reduces unnecessary ER visits.

2. Expanding Rural Access

- AI-mediated care navigation supports underserved areas.

3. Workforce Augmentation

- AI assists overextended clinicians with documentation, triage, and navigation.

4. Payment Modernization

- Contextual partnerships create value-aligned support rather than volume-driven reimbursement.

5. Research and Evaluation

The structured avatar ecosystem allows for measurable outcomes including:

- Time-to-escalation
- Triage accuracy
- Cost reduction metrics
- Chronic care adherence improvements
- Patient satisfaction

VII. Interoperability Considerations

Avatar Tethering can integrate with:

- FHIR-based electronic health record systems
- Telehealth platforms
- Scheduling systems
- Claims and reimbursement infrastructure

Standardized APIs can support scalable deployment across states.

VIII. Governance and Oversight

State implementation frameworks may include:

- Oversight boards
- Provider participation agreements
- Transparency requirements
- Ethical review committees
- Audit capabilities

This ensures accountability while enabling innovation.

IX. Potential National Application

The structured nature of Avatar Tethering makes it adaptable for:

- State-level pilot programs
- CMMI demonstration models
- Public-private partnerships
- Rural Health Transformation initiatives

Its patented architecture offers a standardized but customizable framework for coordinated AI healthcare ecosystems.

X. Conclusion

Avatar Tethering provides a scalable, governance-compatible methodology for integrating AI into clinical care while:

- Preserving licensed provider authority
- Supporting rural hospital sustainability
- Enhancing preventive care delivery
- Encouraging responsible partnership models
- Creating measurable performance metrics

It represents a structured approach to AI deployment that balances innovation, regulation, reimbursement modernization, and rural healthcare preservation.